Projet n°176

Titre : Dynamic of RND and forecasting the price of derivatives

Description : A risk-neutral distribution is a model distribution that makes it possible to calculate a price for a derivative that is consistent with the market price. In the Black-Scholes approach, prices follow a log-normal distribution. It is an example of a risk-neutral distribution but many other risk-neutral probabilities are possible. Moreover, the Black-Scholes model does not apply to all the derivatives on the same underlying, leading to the famous smile of the volatility. Other risk-neutral distributions are designed to explain the price of all these derivatives together. They provide a distribution that is more consistent with the whole market.

In the perspective of derivatives trading or hedging, the knowledge of the dynamic of the risk-neutral distribution can be very helpful. We propose, in this project, to study several risk-neutral distributions, to determine whether they contain some predictive feature on their future evolution, whether they are more or less stable through time. The last step of the project will consist in building a trading strategy based on the hedging of derivatives while taking into account both the forecast of the future risk-neutral distribution (the Greeks are not the same in the forecast distribution) and the uncertainty of this forecast.

1- Collection of data and creation of a database of time series of option prices

2- Calibration of risk-neutral distributions [6]
a. Breeden-Litzenberger approach [5]
b. Fourier approach [3,4]
c. Distortion of historical probability [1]

3- Relative entropy for studying the stability of time series of risk-neutral distributions [2]

4- Modelling the dynamic of risk-neutral distributions
a. Forecasting with a factor model [7]
b. Uncertainty of risk-neutral distributions [8]

5- Trading strategy with models of dynamics of risk-neutral distributions

Expected deliverable: a working code (containing a substantial part programmed in OCaml or Julia) and a thesis written in LaTeX detailing the bibliography and the results obtained.

Bibliography

[1] Avellaneda, M., Buff, R., Friedman, C., Grandechamp, N., Kruk, L., & Newman, J. (2001). Weighted Monte Carlo: a new technique for calibrating asset-pricing models. International Journal of Theoretical and Applied Finance, 4(01), 91-119.

[2] Cover, T. M. & Thomas, J. A. (1991). Elements of information theory. Wiley, New York.

[3] Cui, Z., Liu, Y., & Yao, Y. (2025). Model-free Option-Implied Quantiles. Preprint.

[4] Cui, Z., & Yu, Z. (2021). A model-free Fourier cosine method for estimating the risk-neutral density. The Journal of Derivatives.

[5] Figlewski, S. (2010), Estimating the Implied Risk-Neutral Density for the US Market Portfolio, in Volatility and Time Series Econometrics: Essays in Honor of Robert F. Engle.

[6] Figlewski, S. (2018). Risk-neutral densities: A review. Annual Review of Financial Economics, 10(1), 329-359.

[7] Giacomini, E., Härdle, W., & Krätschmer, V. (2009). Dynamic semiparametric factor models in risk neutral density estimation. AStA Advances in Statistical Analysis, 93(4), 387-402.

[8] Taboga, M. (2016). Option-implied probability distributions: How reliable? How jagged?. International Review of Economics & Finance, 45, 453-469.

Informations sur le projet

Compétences développées : The skills developed are financial, with the pricing and hedging of derivatives as well as the calibration of risk-neutral densities, but also statistical, in order to build dynamic models, and finally computer science, with a significant part of programming.

Informations complémentaires :

Majeure(s) concernée(s) : IF, FIN, FIN_Alt, IF_Alt

Année(s) concernée(s) : A4, A5

Acceptera une équipe A4 si aucune équipe A5 ne se positionne : Oui

Mot(s)-clé(s) concerné(s) : Finance, Finance quantitative, Gestion du risque, Modélisation, Programmation, Séries temporelles, Statistiques, Stratégies systématiques

Informations sur le partenaire

Entreprise / Association / École : ESILV

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Prénom : Matthieu